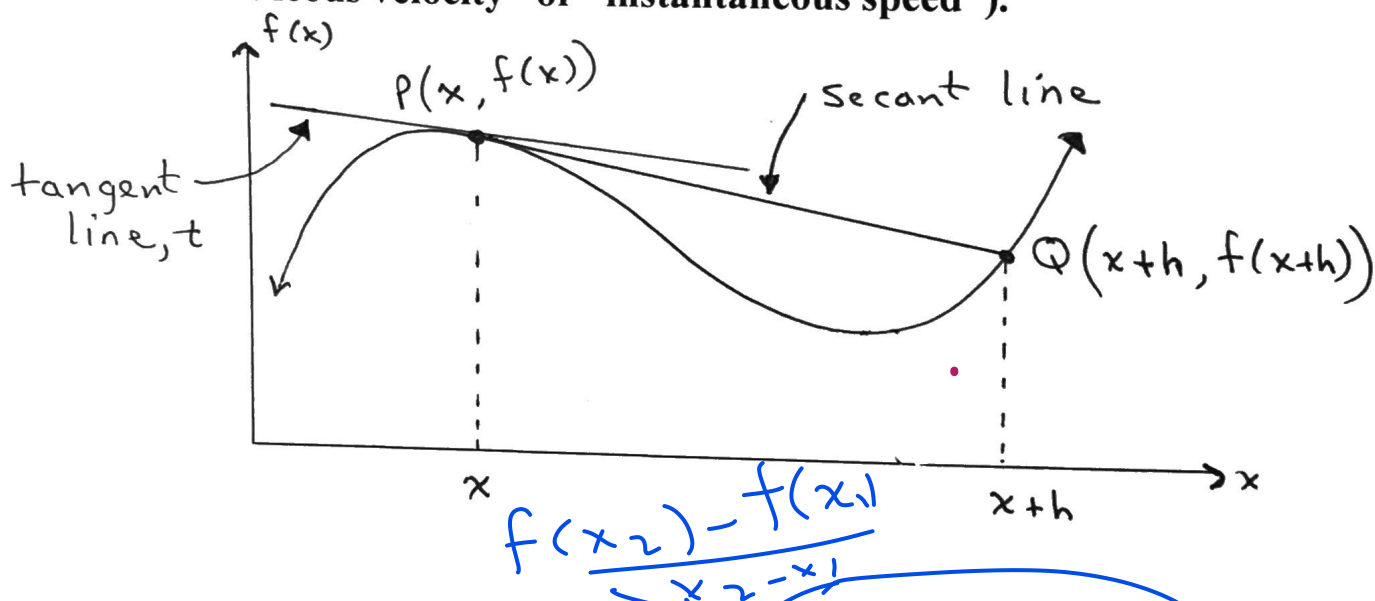


Slopes of Secants and Tangents

- The slope of the secant is also the average rate of change (AROC). (From an application point of view it is often referred to “average velocity” or “average speed”).
- The slope of the tangent is also the instantaneous rate of change (IROC). (From an application point of view it is often referred to “instantaneous velocity” or “instantaneous speed”).



Slope of the secant line PQ is given by

$$m_{PQ} = \frac{f(x+h) - f(x)}{h}$$

Sometimes called the “Newton Quotient” or the “difference quotient”.

Slope of the tangent line is given by

$$m_t = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

IROC

Example 1: Determine the slope of the tangent to the curve

$f(x) = x^2 + 1$ at the point where $x = 2$.

slope of
tangent

$$m_t = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

1st principles
definition

$$m_t = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 1 - (x^2 + 1)}{h}$$

$$m_t = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 1 - x^2 - 1}{h}$$

$$m_t = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h}$$

$$\therefore m_t = 2x$$

$$m_t|_{x=2} = 4$$

$$m_t = 2x$$

Example 2: Determine the equation of the tangent to the curve

$f(x) = \sqrt{x+3}$ at the point where $x = 1$.

$$m_t = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$m_t = \lim_{h \rightarrow 0} \frac{\sqrt{x+h+3} - \sqrt{x+3}}{h}$$

foil

$$m_t = \lim_{h \rightarrow 0} \left(\frac{\sqrt{x+h+3} - \sqrt{x+3}}{h} \right) \left(\frac{\sqrt{x+h+3} + \sqrt{x+3}}{\sqrt{x+h+3} + \sqrt{x+3}} \right)$$

$$m_t = \lim_{h \rightarrow 0} \frac{x+h+3 - (x+3)}{h(\sqrt{x+h+3} + \sqrt{x+3})}$$

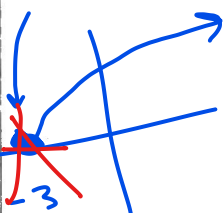
$$m_t = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h+3} + \sqrt{x+3}}$$

$$\therefore m_t|_{x=1} = \frac{1}{4}$$

Equation:

$$y - y_1 = m(x - x_1)$$

$$y - 2 = \frac{1}{4}(x - 1)$$



No
unique
tangent

$x = -3$
 m_t is undefined

$$m_t = \frac{1}{2\sqrt{x+3}}$$

$$f(1) = 2$$

slope-y-int form
 $y = mx + b$

standard form
 $Ax + By + C = 0$

point-slope form
 $y - y_1 = m(x - x_1)$

Example 3: At what point on the curve $f(x) = -2x^4$ is the tangent line perpendicular to the line $x - y + 1 = 0$?

Note: the line that is perpendicular to the tangent line at the point of tangency, P is called the "normal line".

* $x - y + 1 = 0 \Rightarrow y = x + 1 \quad \therefore m = 1$; $m_t = -1$
(negative reciprocal)

* $m_t = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$m_t = \lim_{h \rightarrow 0} \frac{-2(x+h)^4 + 2x^4}{h}$

$m_t = \lim_{h \rightarrow 0} \frac{-2[(x+h)^4 - x^4]}{h}$

$m_t = \lim_{h \rightarrow 0} \frac{-2[(x+h)^2 - x^2][(x+h)^2 + x^2]}{h}$

$m_t = \lim_{h \rightarrow 0} \frac{-2[\cancel{x^2} + 2xh + \cancel{h^2} - \cancel{x^2}][(x+h)^2 + x^2]}{h}$

$m_t = \lim_{h \rightarrow 0} \frac{-2[2x+h][(x+h)^2 + x^2]}{h}$

$m_t = \lim_{h \rightarrow 0} -2[2x+h][(x+h)^2 + x^2]$

$m_t = -8x^3$

But $m_t = -1 \Rightarrow -8x^3 = -1$

$x^3 = \frac{1}{8}$

$x = \frac{1}{2}$

could use
Pascal's Δ .

$$\begin{array}{r} 1 \\ 1 \ 1 \\ 1 \ 4 \ 1 \\ 1 \ 3 \ 3 \ 1 \\ 1 \ 6 \ 6 \ 4 \ 1 \end{array}$$

$\lim_{x \rightarrow a} f(x)$

$= c \lim_{x \rightarrow a} f(x)$

$\therefore$ point is

$\left(\frac{1}{2}, -\frac{1}{8}\right)$

$f\left(\frac{1}{2}\right)$